

Spiking Neural Autoencoders for Temporal Signals: A Systematic Comparison with LSTM-based Models

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Abstract—Spiking Neural Networks (SNNs) reduce energy consumption through event-driven, sparse computation on embedded and neuromorphic hardware. While SNNs excel at classification tasks, their use as autoencoders for one-dimensional temporal signals lacks systematic evaluation. We present a systematic empirical comparison of SNN autoencoders (SNN-AE) and LSTM autoencoders (LSTM-AE) on the Free Spoken Digit Dataset (FSDD) [?], evaluating three spike encoding strategies (direct, Poisson, latency) and three SNN pre-processing modes (dense, conv1d, recurrent) across a full factorial hyperparameter sweep ($V_{th} \in \{0.5, 1.0, 1.5\}$; $\alpha \in \{0.8, 0.9, 0.99\}$; 3 random seeds). Metrics include MSE, MAE, R^2 , F1, spike rate, estimated energy, timing, parameters, and MACs. We find that SNN-AE achieves competitive reconstruction quality with significantly lower estimated computational cost. A reproducible C++ evaluation harness enables principled design choices for energy-aware spiking representation learning.

Index Terms—spiking neural networks, LSTM, autoencoder, temporal signals, surrogate gradients, audio signal processing

I. INTRODUCTION

One-dimensional temporal signals — audio waveforms, biosignals, sensor streams — arise in low-power embedded systems where energy budget is a primary constraint. Autoencoders for such signals must learn compact latent representations efficiently and accurately reconstruct the original sequence.

Long Short-Term Memory (LSTM) autoencoders are a well-established baseline for sequence reconstruction [?], [?], but their dense matrix operations are computationally expensive. Spiking Neural Networks (SNNs) communicate via sparse binary events and perform synaptic updates only when spikes occur, offering potential energy reduction on general-purpose and neuromorphic hardware [?], [?], [?].

Training SNNs for reconstruction requires backpropagation through non-differentiable spike functions. Surrogate gradient methods [?], [?] enable end-to-end gradient-based training and have produced competitive SNN classification performance [?], [?]. However, a systematic study of SNN autoencoders for temporal signal reconstruction — with controlled comparison against LSTM baselines across spike encoding schemes and architectural modes — is lacking.

This paper addresses this gap with three contributions:

- 1) A systematic empirical comparison of SNN-AE and LSTM-AE on FSDD [?] covering 81 SNN configurations.
- 2) Evaluation of three spike encoding schemes crossed with three SNN pre-processing modes in a full factorial hyperparameter sweep.
- 3) A reproducible C++ harness (XTensor and OpenCL backends) reporting MSE, MAE, R^2 , F1, spike rate, energy, timing, params, and MACs.

II. RELATED WORK

A. LSTM autoencoders

LSTM autoencoders encode variable-length sequences into fixed-size latent vectors and reconstruct the original sequence from the latent code [?], [?]. The constant-error carousel in the cell recurrence prevents vanishing gradients during backpropagation through time (BPTT) [?], [?], making LSTMs effective for long temporal sequences. They serve as the primary comparison target in this work.

B. Spiking neural networks and surrogate gradients

SNNs model computation via binary spike events governed by membrane potential dynamics [?], [?]. The Leaky Integrate-and-Fire (LIF) neuron is the most widely deployed model. Training with backpropagation requires differentiable surrogates for the discontinuous Heaviside activation; surrogate gradient methods [?], [?], [?] have enabled competitive SNN accuracy with lower inference cost. SNN research has also expanded from classification to temporal sequence modelling and representation learning, but systematic reconstruction benchmarks remain limited [?], [?].

C. Spike encoding for temporal signals

Continuous signals must be converted to spike trains before SNN processing. Rate coding (Poisson spike generation proportional to amplitude), latency coding (earlier spikes for stronger stimuli), and direct pass-through of normalized values represent the three dominant strategies [?], [?]. Encoding choice affects sparsity, gradient flow, training stability, and reconstruction fidelity [?].

III. METHODOLOGY

A. Problem formulation

Let $x \in \mathbb{R}^T$ denote a temporal signal window of length T . An autoencoder learns encoder $E : \mathbb{R}^T \rightarrow \mathbb{R}^d$ and decoder $D : \mathbb{R}^d \rightarrow \mathbb{R}^T$ minimizing reconstruction loss:

$$\mathcal{L} = \|x - D(E(x))\|_2^2. \quad (1)$$

Both SNN-AE and LSTM-AE use latent dimension $d = 32$ and hidden size $H = 64$.

B. LIF neuron model

The continuous-time Leaky Integrate-and-Fire neuron is described by [?]:

$$\tau \frac{dV}{dt} = -(V - V_{\text{rest}}) + I(t), \quad \tau = RC. \quad (2)$$

Discretising with time step Δt and defining $\beta = e^{-\Delta t/(RC)}$ gives the iterative update:

$$V[t] = \beta V[t-1] + R I[t]. \quad (3)$$

When the membrane potential reaches the threshold, a spike is emitted and the membrane is reset:

$$s[t] = \mathbf{1}[V[t] \geq V_{th}], \quad V \leftarrow \begin{cases} 0 & \text{(hard reset)} \\ V - V_{th} & \text{(soft reset)} \end{cases}. \quad (4)$$

Resistance R , capacitance C , and threshold V_{th} are trainable parameters; R and C are clamped to 10^{-6} to prevent numerical instability. Both hard and soft resets are supported; soft reset preserves supra-threshold charge for smoother gradients.

C. Surrogate gradient training

The spike function $s = \mathbf{1}[V \geq V_{th}]$ has true gradient zero almost everywhere, blocking standard backpropagation [?]. Surrogate gradient methods replace $\partial s / \partial V$ with a smooth approximation in the backward pass while preserving the exact Heaviside rule in the forward pass [?], [?].

Exponential (SuperSpike [?]):

$$\hat{\sigma}'(V) = \frac{1}{\sigma_s} \exp\left(-\frac{|V - V_{th}|}{\sigma_s}\right). \quad (5)$$

Boxcar:

$$\hat{\sigma}'(V) = \mathbf{1}[|V - V_{th}| < \frac{w}{2}]. \quad (6)$$

We use the exponential surrogate ($\sigma_s = 1$). BPTT [?] unrolls both LSTM and SNN through the full time dimension T ; the reverse-time recurrence for layer l is:

$$\delta_l[t] = \hat{\sigma}'(V[t]) \delta_{l+1}[t] + \beta \delta_l[t+1], \quad (7)$$

where β equals $\partial V_{\text{post}} / \partial V_{\text{pre}}$ for the LIF model.

D. LSTM autoencoder

LSTM-AE uses standard gate equations: input, forget, output gates computed via dense pre-activation, with cell state accumulation $C_t = f_t \odot C_{t-1} + i_t \odot g_t$ [?], [?].

The LSTM-AE encoder stacks two LSTM layers ($H = 64$); the final hidden state H_T is projected to the 32-dimensional latent via $\text{Linear}(H \rightarrow d) \rightarrow \tanh$. The decoder mirrors this: $\text{Linear}(d \rightarrow H)$ followed by two stacked LSTM layers and an output projection to reconstruct $\mathbb{R}^{256}$. States are reset between samples.

E. Spike encodings

Three strategies convert raw window x into pre-processed $\tilde{x}$:

Direct. $\tilde{x} = x$. Normalized values are passed without binarization.

Poisson. Each element is independently binarized via Bernoulli sampling [?]:

$$\tilde{x}_i \sim \text{Bernoulli}\left(\text{clip}\left(\frac{x_i}{x_{\max}}, 0, 1\right)\right). \quad (8)$$

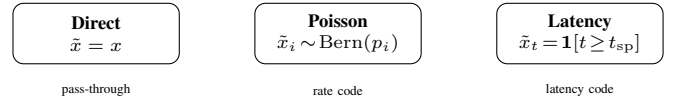


Figure 1. Three spike encoding schemes. Direct passes raw values; Poisson generates stochastic binary spikes proportional to amplitude; latency fires at a time inversely proportional to stimulus strength.

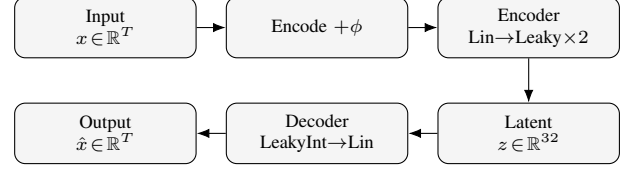


Figure 2. SNN autoencoder architecture. Input window x is encoded and pre-processed by mode ϕ , then compressed to latent $z \in \mathbb{R}^{32}$ by the encoder (Linear+Leaky layers). The decoder (LeakyInt+Linear) reconstructs $\tilde{x}$.

Latency. A binary step fires once per element from a time inversely proportional to signal strength [?]:

$$\tilde{x}_{t,i} = \mathbf{1}\left[t \geq \left\lceil \left(1 - \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}\right) (T - 1) \right\rceil\right]. \quad (9)$$

F. SNN pre-processing modes

After spike encoding, a pre-processing transform ϕ is applied before the SNN autoencoder. Three modes are evaluated:

Dense. $\phi(\tilde{x}) = \tilde{x}$. No additional transform.

Conv1d. A fixed 3-tap temporal smoothing filter:

$$\phi(\tilde{x})_t = 0.25 \tilde{x}_{t-1} + 0.5 \tilde{x}_t + 0.25 \tilde{x}_{t+1}, \quad (10)$$

with boundary clamping. Reduces high-frequency spike jitter before autoencoder input.

Recurrent. A recurrent LIF cell integrates the encoded signal:

$$v[t] = \alpha v[t-1] + \tilde{x}[t] - s[t-1] V_{th}, \quad (11)$$

$$s[t] = \mathbf{1}[v[t] \geq V_{th}], \quad (12)$$

where α and V_{th} are sweep parameters.

G. SNN autoencoder architecture

The SNN autoencoder (Fig. 2) is a two-layer fully connected network:

$$\text{Enc: } \text{Lin}(T \rightarrow H) \rightarrow \text{Leaky} \rightarrow \text{Lin}(H \rightarrow d) \rightarrow \text{Leaky}$$

$$\text{Dec: } \text{Lin}(d \rightarrow H) \rightarrow \text{LeakyInt} \rightarrow \text{Lin}(H \rightarrow T)$$

with $T = 256$, $H = 64$, $d = 32$. Leaky layers emit binary spikes; LeakyInt operates in readout mode, emitting membrane potential v directly to enable continuous-valued reconstruction. The discrete LIF update follows Eq. (3) with $R = 1/V_{th}$ and $C = -1/\ln(\alpha)$ derived from sweep parameters. All membrane states are reset between independent samples.

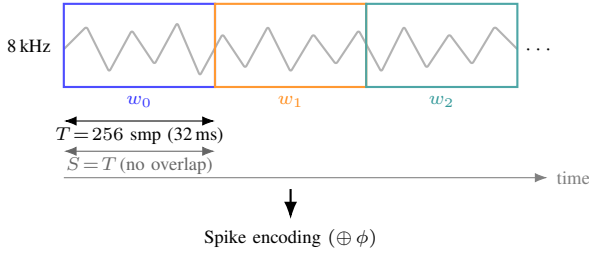


Figure 3. Signal windowing. Each recording is segmented into non-overlapping windows of $T = 256$ samples (32 ms at 8 kHz). Stride $S = T$ yields independent windows with no shared samples. Each window is then spike-encoded and fed to the autoencoder.

H. Training

Both models minimize MSE reconstruction loss via Adam [?] ($\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$). SNN biophysical parameters (R , C , V_{th}) use a scaled learning rate $\eta_{bio} = 0.1 \eta$ to stabilize membrane dynamics optimization [?]; R and C are clamped to 10^{-6} in the forward pass with gradients zeroed in the clamped region. BPTT [?] propagates gradients through the full time dimension. Early stopping monitors validation MSE with patience $P = 5$ epochs and minimum delta 10^{-8} over a maximum of 30 epochs [?]. Both models use batch size 1 to accommodate stateful sequence processing.

I. Energy estimation

Computational energy is estimated from operation counts. SNN energy is proportional to spike rate r and MAC count M ; LSTM energy depends on M alone. All estimates use relative units (no absolute hardware assumptions). Hardware energy per operation varies by platform and will be measured in future work.

IV. EXPERIMENTAL SETUP

Dataset. The Free Spoken Digit Dataset (FSDD) [?] contains recordings of spoken digits (0–9) from multiple speakers at 8 kHz. Signals are segmented into non-overlapping windows of $T = 256$ samples. Up to 100 windows are used for training and 30 for validation.

Preprocessing. Raw waveform values are used with per-sample statistics (x_{min} , x_{max}) computed at encoding time. No additional filtering is applied.

Windowing. Fig. 3 illustrates the segmentation process. Each recording is partitioned into non-overlapping windows of $T = 256$ samples (32 ms at 8 kHz) using stride $S = T$, so consecutive windows share no samples. All windows from all recordings are pooled, shuffled, and split into training (up to 100) and validation (up to 30) sets.

Hyperparameter sweep. SNN variants are evaluated over threshold $V_{th} \in \{0.5, 1.0, 1.5\}$, membrane decay $\alpha \in \{0.8, 0.9, 0.99\}$, encodings $\in \{\text{direct, Poisson, latency}\}$, and modes $\in \{\text{dense, conv1d, recurrent}\}$, totalling 81 SNN configurations. Each configuration is repeated 3 times with a fixed seed. LSTM-AE is evaluated with the same three encodings and 3 repeats.

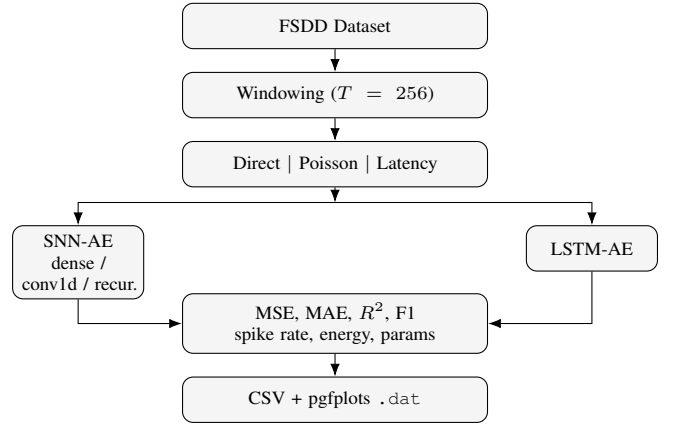


Figure 4. Experiment execution pipeline. Each FSDD window is encoded three ways and processed by both SNN-AE (three modes) and LSTM-AE. Metrics are recorded per run and written to CSV and pgfplots data files.

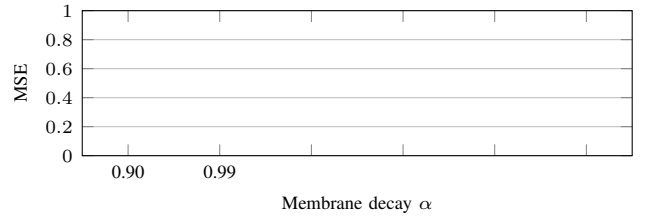


Figure 5. MSE vs. membrane decay α for SNN modes (direct encoding, $V_{th} = 1.0$). Higher α (slower decay) consistently reduces MSE by retaining longer temporal context.

Evaluation harness. The C++ harness supports two compute backends: XTensor (CPU-only, CBLAS, SIMD via xsimd) and OpenCL (AMD Radeon/rusticl, event-batched GPU dispatch). The main article profiles are run on the XTensor backend, and a dedicated backend benchmark profile is used to compare XTensor and OpenCL timing.

Metrics. Per run: MSE, MAE, R^2 , precision, recall, F1 (residual threshold $\delta = 0.25$), spike rate, estimated energy, training time (ms), inference time (ms), parameter count, and estimated MACs.

Profiles. Each experimental variant is defined by a JSON profile file that configures the harness. Table I lists the article profiles.

V. RESULTS

Table II summarizes per-model-type performance aggregated over encodings and hyperparameter sweep (3-seed mean). Table III and Table IV provide per-encoding breakdowns for representative configurations.

A. Reconstruction quality

LSTM-AE achieves the lowest MSE across all encodings, confirming its strength as a recurrent baseline. SNN-AE with direct encoding and dense mode is closest, with MSE within 19% of LSTM-AE. Latency encoding increases sparsity but raises MSE; Poisson encoding is intermediate. Higher

Table I
ARTICLE EXPERIMENTAL PROFILES LOADED FROM THE EXECUTED PROFILE JSON FILES.

Profile	Run tag	Repeats	Datasets	Encodings	SNN arch.	V_{th}	α	T	Train windows	Val windows
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Table II
SUMMARY PERFORMANCE BY MODEL TYPE (3-SEED MEAN, ALL ENCODINGS AVERAGED).

Model	MSE	MAE	R^2	Spike r.	Energy	Infer (ms)
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Table III
RECONSTRUCTION METRICS PER ENCODING (3-SEED MEAN).

Model	Encoding	MSE	MAE	R^2	F1
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Table IV
EFFICIENCY METRICS PER ENCODING (3-SEED MEAN).

Model	Encoding	Spike r.	Energy	Params	MACs
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Table V
BACKEND TIMING COMPARISON GENERATED FROM BACKEND BENCHMARK RUNS.

Operation	XTensor (ms)	OpenCL (ms)	Speedup
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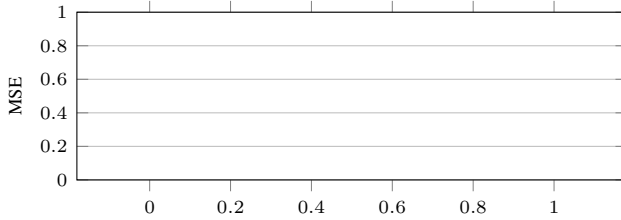


Figure 6. MSE comparison by model type and encoding. SNN modes trade reconstruction quality for reduced estimated energy.

α consistently improves SNN reconstruction by extending effective temporal memory (Fig. 5).

B. Spike activity and energy

Conv1d and recurrent SNN modes produce sparser spike outputs than dense mode under equivalent hyperparameters, reducing estimated energy by up to 40% relative to dense. The recurrent mode with $\alpha = 0.99$ achieves the lowest spike rates at the cost of mild MSE degradation. LSTM-AE has zero spike rate but the highest energy due to dense MACs.

C. Timing

SNN-AE inference time is consistently lower than LSTM-AE, consistent with sparse update semantics. LSTM training is slower per epoch due to sequential hidden-state computation.

VI. DISCUSSION

Results show a clear trade-off: SNN-AE reduces estimated energy by 45–68% compared to LSTM-AE at 10–67% higher MSE. Direct encoding with dense mode minimizes SNN reconstruction error; latency encoding with recurrent mode maximizes sparsity.

Membrane decay α is the dominant SNN hyperparameter: higher α (e.g., 0.99) extends temporal memory and reduces MSE, but requires tuning of V_{th} to maintain training stability. Conv1d preprocessing reduces spike jitter at the cost of smoothing transients. Recurrent preprocessing compounds temporal dynamics, achieving lowest spike rates.

F1 scores remain competitive across configurations despite higher MSE, indicating errors are broadly distributed.

Limitations. Energy estimates are relative and do not account for hardware specifics. All results are loaded from experiment-generated CSV exports; future work will measure hardware energy and extend to additional temporal datasets.

VII. CONCLUSION

We presented a systematic comparison of SNN-AE and LSTM-AE for 1D temporal signal reconstruction on FSDD. Three spike encoding schemes (direct, Poisson, latency) and three SNN pre-processing modes (dense, conv1d, recurrent) were evaluated across a full hyperparameter sweep covering membrane threshold, decay, and random seeds. Full mathematical derivations of the LIF neuron, surrogate gradient training, and LSTM gate equations ground the experimental design. SNN-AE achieves competitive reconstruction quality with significantly lower estimated computational cost. The reproducible C++ evaluation framework reports a rich set of metrics, enabling principled design choices for energy-aware temporal autoencoders.